as **RNA dependent DNA polymerase or transcriptase** which can transcribe RNA into DNA.

Resistance

Viruses are **inactivated** by sunlight, UV rays and ionizing radiations. The most active **antiviral disinfectants** are oxidizing agents such as hydrogen peroxide, potassium permanganate and hypochlorites. Organic iodine compounds are actively **virucidal**. Chlorination of drinking water kills most viruses but its efficacy is influenced by the presence of organic matter. Some viruses such as hepatitis virus, polio viruses are relatively resistant to chlorination.



Antibiotics active against bacteria are completely ineffective against viruses.

Viral Multiplication

The genetic information necessary for viral replication is contained in the viral nucleic acid, and also depends on the synthetic machinery of the host cell for replication. The Viral multiplication cycle can be divided into **six steps** and they are as follows, **1. Adsorption or attachment, 2. Penetration, 3. Uncoating, 4. Bio synthesis, 5. Maturation** and **6. Release.**

1. Adsorption

Virions may come into contact with cells by random collision but adsorption takes place only if there is an **affinity** between the **virus and the host**. The cell surface should contain specific receptor site for the virus to attach on to it.

2. Penetration

Bacteria possess rigid cell walls, only the viral nucleic acid is introduced intracellularly by a complex mechanism. Animal cells do not have rigid cell walls and the whole virus can enter and virus particles may be engulfed by a mechanism resembling **phagocytosis**, a process known as **'Viropexis'**. In case of the **enveloped viruses**, the viral envelope may **fuse** with the plasma membrane of the host cell and release the nucleocapsid into the cytoplasm.

3. Uncoating

Release of the viral nucleic acid from the capsid into the host cell with most viruses, uncoating is affected by the action of lysosomal enzymes of the host cell.

4. Biosynthesis

Virus can synthesise viral nucleic acid, capsid protein and also the enzymes necessary in the various stages of viral synthesis, assembly and release. In addition certain regulator proteins are also synthesized. Most **DNA viruses** synthesise their nucleic acid in the **host cell nucleus**. Most **RNA viruses** synthesise all their components in the cytoplasm.

Biosynthesis consists of the following steps:

i. Transcription of messenger RNA (mRNA) from the viral nucleic acid.

ii. Translation of the mRNA into 'early proteins'

iii. Replication of Viral nucleic acid.

iv. Synthesis of late or structural proteins, which are the components of daughter virion capsids.

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